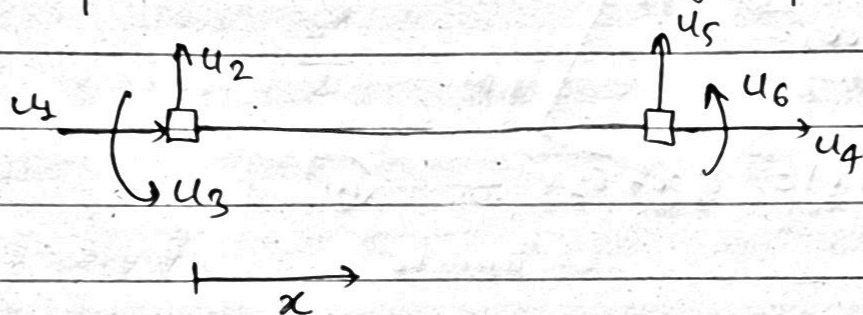


Homework: 1

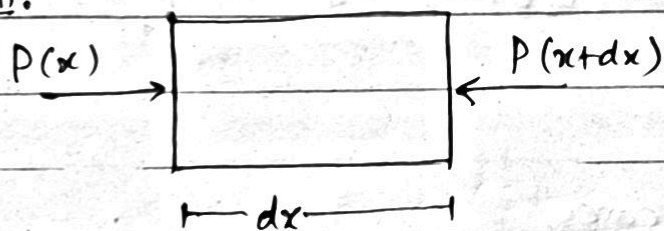
1

- ① For the two-node linear-elastic Euler-Bernoulli beam element shown below with both axial and bending degrees of freedom, derive all the components of the 6x6 element stiffness matrix. Follow the process reviewed in the class and develop the governing differential equations for axial and bending - by writing the equilibrium, compatibility, and constitutive equations and then solve for proper boundary conditions.



→ Solⁿ

- ② For axial degrees of freedom,
Assuming no distributed axial load,
Equilibrium:



$$\frac{dP(x)}{dx} = 0 \quad \text{--- ①}$$

Compatibility: strain-displacement relationship

$$\epsilon_x = \frac{du}{dx}$$

Constitutive:

for linear-elastic material,
from Hooke's law,

$$\sigma_x = E \epsilon_x \quad \Rightarrow \quad P = EA \epsilon_x = EA \frac{du}{dx} \quad \text{①}$$

from eqⁿ (I),

$$\frac{d}{dx} \left(EA \frac{du}{dx} \right) = 0$$

$$\Rightarrow EA \frac{d^2u}{dx^2} = 0 \quad \text{--- (II)}$$

Integrating twice,

$$u(x) = C_1 x + C_2 \quad \text{--- (III)}$$

Boundary Conditions

$$\text{At } x=0, \quad u(0) = u_1 \Rightarrow C_2 = u_1$$

$$\text{At } x=L, \quad u(L) = u_2 \Rightarrow C_1 L + u_1 = u_2 \Rightarrow C_1 = \frac{u_2 - u_1}{L}$$

Thus eqⁿ (III) becomes,

$$u(x) = \left(1 - \frac{x}{L}\right) u_1 + \left(\frac{x}{L}\right) u_2 \quad \text{--- (IV)}$$

from eqⁿ (I),

$$P(x) = EA \frac{du}{dx} = \frac{EA}{L} (u_2 - u_1)$$

At $x=0$, the external force must balance the internal force pointing away from the boundary,

$$F_{1x} = -P(0) = -\frac{EA}{L} (u_2 - u_1) = \frac{EA}{L} u_1 - \frac{EA}{L} u_2$$

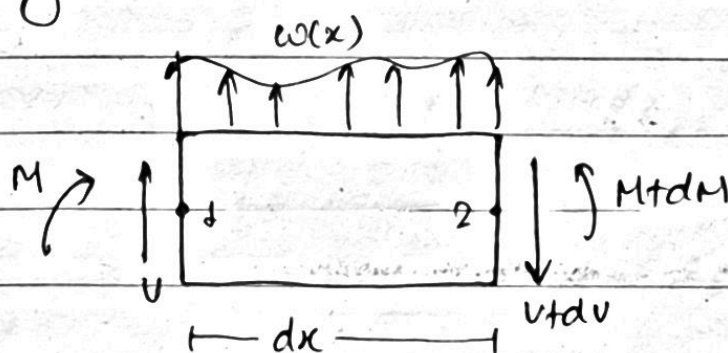
At $x=L$, the external force acts in the direction of internal tension,

$$\therefore F_{qx} = P(L) = -\frac{EA}{L} u_2 + \frac{EA}{L} u_4$$

In matrix form,

$$[K]_{axial} = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{--- (1a)}$$

(b) For bending DOFs,



Equilibrium:

$$\sum F_y = 0 \Rightarrow V - (V+dv) + w(x) dx = 0$$

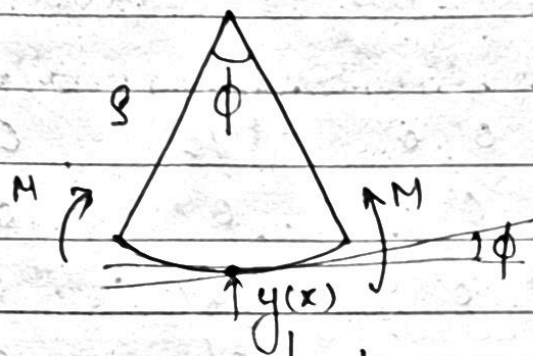
$$\Rightarrow \frac{dV}{dx} = w(x) \quad \text{--- (V)}$$

$$\sum M_2 = 0 \Rightarrow V dx + M - (M+dm) + w(x) dx \left(\frac{dx}{2} \right) = 0$$

$$\Rightarrow \frac{dM}{dx} = V \quad \text{--- (VI)}$$

Constitutive:

$$\text{Curvature } (k) = \frac{1}{\rho} = \frac{M}{EI} \quad \text{--- (VII)}$$



transverse displacement function y

Compatibility:

for small angle/slope,

$$\tan \phi \approx \phi \approx \frac{dy}{dx}$$

$$\Rightarrow \phi = \frac{dy}{dx}$$

$$\text{rate of change of slope} = k = \frac{d^2y/dx^2}{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}$$

$$\approx \frac{d^2y}{dx^2} = \frac{d\phi}{dx} \quad \text{--- (VIII)}$$

from (VII) and (VIII),

$$\frac{d^2y}{dx^2} = \frac{M}{EI} \quad \Rightarrow \quad M = EI \frac{d^2y}{dx^2} \quad \text{--- (IX)}$$

And, from (v), (vi), (ix),

$$\frac{d^2 M}{dx^2} = w(x) = \frac{d^2}{dx^2} \left(EI \frac{d^2 y}{dx^2} \right)$$

$$\frac{dM}{dx} = V = \frac{d}{dx} \left(EI \frac{d^2 y}{dx^2} \right) \quad \text{--- (x)}$$

for constant EI , $V = EI \frac{d^3 y}{dx^3}$ and if only nodal forces and

moments exist, i.e. $w(x) = 0 \Rightarrow \frac{d^4 y}{dx^4} = 0$ --- (xi)

Integrating eqⁿ (xi) four times,

$$y(x) = a_4 x^3 + a_3 x^2 + a_2 x + a_1 \quad \text{--- (xii)}$$

$$\Rightarrow \phi(x) = 3a_4 x^2 + 2a_3 x + a_2 \quad \text{--- (xiii)}$$

Boundary Conditions,

$$\text{At } x=0, y(0) = u_2 \Rightarrow a_1 = u_2 \quad \text{--- (xiv)}$$

$$\text{At } x=0, \phi(0) = u_3 \Rightarrow a_2 = u_3 \quad \text{--- (xv)}$$

$$\text{At } x=L, y(L) = u_5 \Rightarrow a_4 L^3 + a_3 L^2 + a_2 L + a_1 = u_5 \quad \text{--- (xvi)}$$

$$\text{At } x=L, \phi(L) = u_6 \Rightarrow 3a_4 L^2 + 2a_3 L + a_2 = u_6 \quad \text{--- (xvii)}$$

In matrix form,

$$\{d\} = \begin{Bmatrix} u_2 \\ u_3 \\ u_5 \\ u_6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ l^3 & l^2 & l & 1 \\ 3l^2 & 2l & 1 & 0 \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix} \quad \text{--- (XVIII)}$$

$$y = [N_1 \ N_2 \ N_3 \ N_4] \{d\} = [N] \{d\}$$

where, $[N]$ = Shape functions or interpolation functions known as Hermite cubic interpolation functions

Using eqⁿ (XII), (XVIII),

$$N_1 = \frac{1}{l^3} (2x^3 - 3x^2l + l^3)$$

$$N_2 = \frac{1}{l^3} (x^3l - 2x^2l^2 + x l^3)$$

$$N_3 = \frac{1}{l^3} (-2x^3 + 3x^2l)$$

$$N_4 = \frac{1}{l^3} (x^3l - x^2l^2)$$

Using eqⁿ (IX) and (X),

$$f_{xy} = V = EI \frac{d^3(y(x))}{dx^3} = \frac{EI}{l^3} (12u_2 + 6l u_3 - 12u_5 + 6l u_6) \quad \text{--- (2)}$$

$$m_1 = -m = -EI \frac{d^2 y(0)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 4L^2 u_3 - 6Lu_5 + 2L^2 u_6) \quad (3)$$

$$f_{1y} = -V = -EI \frac{d^3 y(L)}{dx^3} = \frac{EI}{L^3} (-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6) \quad (4)$$

$$m_2 = m = EI \frac{d^2 y(L)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 2L^2 u_3 - 6Lu_5 + 4L^2 u_6) \quad (5)$$

Using (1a), (2), (3), (4), (5),

$$\begin{Bmatrix} f_{1x} \\ f_{1y} \\ m_1 \\ f_{2x} \\ f_{2y} \\ m_2 \end{Bmatrix} = \begin{bmatrix} AE/L & 0 & 0 & -AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix}$$

$$\equiv \{F\} = [K] \{q\}$$

Stiffness Matrix is symmetric because,

→ Work done by force A moving through displacement B equals the work done by force B moving through displacement A

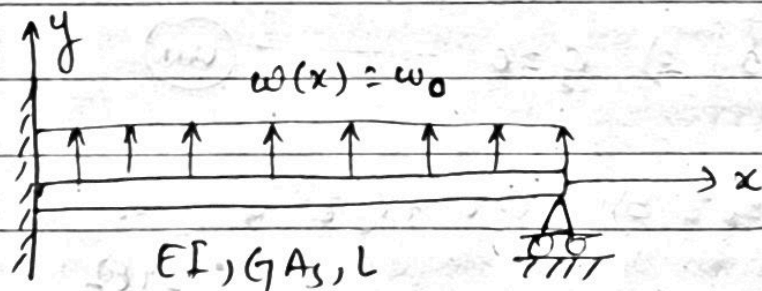
→ Proof by Betti's Reciprocal Theorem

$$F_1 \cdot u_{21} = F_2 \cdot u_{12} \Rightarrow K_{ij} = K_{ji}$$

→ In inelastic systems, the stiffness matrix is generally not symmetric because the work done is not same.

- ② Develop and solve analytically the governing differential equation for the following beam considering three cases: (a) Euler-Bernoulli beam theory, (b) pure shear beam theory, and (c) Timoshenko beam theory. For each case, derive equations for the full displacement/deformation field including vertical deformation $y(x)$, slope $\theta(x)$, and shear deformation $\gamma(x)$, and internal forces including $V(x)$ and $M(x)$.

Observe that the solution derived for the Timoshenko beam will be equivalent to the Euler-Bernoulli beam assuming $GA_s \rightarrow \infty$, and equivalent to the pure shear beam assuming $EI \rightarrow \infty$.



→ Solⁿ. (Ignoring Axial Deformation)

(a) Euler-Bernoulli Beam Theory
from problem #1,
for constant EI ,

$$EI \frac{d^4 y}{dx^4} = w(x) \quad \text{--- (I)}$$

$$M = EI \frac{d^2 y}{dx^2} \quad \text{--- (II)}$$

$$V = EI \frac{d^3 y}{dx^3} \quad \text{--- (III)}$$

$$\theta = \frac{dy}{dx} \quad \text{--- (IV)}$$

Integrating eqⁿ (I) four times,

$$y(x) = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \frac{w_0}{24EI} x^4 \quad \text{--- (V)}$$

$$\alpha(x) = \frac{dy}{dx} = C_1 + 2C_2 x + 3C_3 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (vi)}$$

Boundary Conditions,

$$\text{At } x=0, y(0)=0 \Rightarrow C_0=0 \quad \text{--- (vii)}$$

$$\text{At } x=0, \alpha(0)=0 \Rightarrow C_1=0 \quad \text{--- (viii)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow C_2 L^2 + C_3 L^3 + \frac{w_0}{24EI} L^4 = 0 \quad \text{--- (ix)}$$

$$\text{At } x=L, M(L)=0 \Rightarrow EI \left(2C_2 + 6C_3 L + \frac{w_0}{2EI} L^2 \right) = 0$$

$$\Rightarrow 2C_2 + 6C_3 L + \frac{w_0}{2EI} L^2 = 0 \quad \text{--- (x)}$$

from (ix) and (x),

$$\left(-3C_3 L - \frac{w_0}{4EI} L^2 \right) L^2 + C_3 L^3 + \frac{w_0}{24EI} L^4 = 0$$

$$\Rightarrow C_3 L = -\frac{w_0 L^2}{24EI} + 3C_3 L + \frac{w_0}{4EI} L^2$$

$$\Rightarrow 2C_3 L = \frac{w_0 L^2}{EI} \left(\frac{1}{24} - \frac{1}{4} \right) \Rightarrow C_3 = -\frac{5}{48} \frac{w_0 L}{EI} \quad \text{--- (xi)}$$

$$\begin{aligned}
 \therefore C_2 &= -3L \left(\frac{-5}{48} \frac{\omega_0 L}{EI} \right) - \frac{\omega_0}{4EI} L^2 \\
 &= \frac{15}{48} \frac{\omega_0 L^2}{EI} - \frac{\omega_0 L^2}{4EI} \\
 &= \frac{1}{16} \frac{\omega_0 L^2}{EI} \quad \text{--- (xii)}
 \end{aligned}$$

Thus,

$$y(x) = \frac{1}{16} \frac{\omega_0 L^2}{EI} x^2 - \frac{5}{48} \frac{\omega_0 L}{EI} x^3 + \frac{\omega_0}{24EI} x^4 \quad \text{--- (1a)}$$

$$\alpha(x) = \frac{1}{8} \frac{\omega_0 L^2}{EI} x - \frac{5}{16} \frac{\omega_0 L}{EI} x^2 + \frac{\omega_0}{6EI} x^3 \quad \text{--- (2a)}$$

$$\gamma(x) = 0 \quad \text{--- (3a)}$$

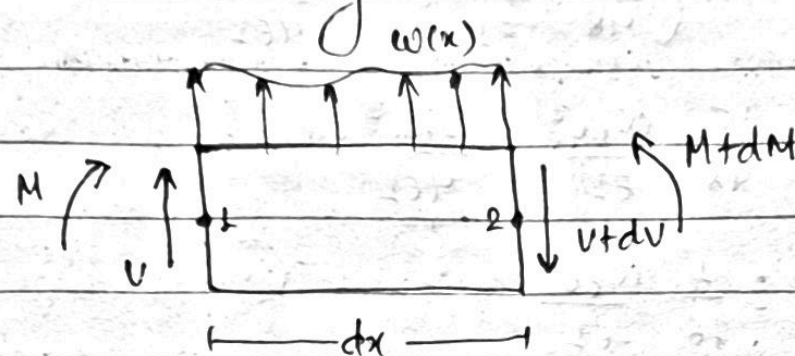
$$M(x) = EI \left(2C_2 + 6C_3 x + \frac{\omega_0}{2EI} x^2 \right)$$

$$\Rightarrow M(x) = \frac{\omega_0 L^2}{8} - \frac{5}{8} \omega_0 L x + \frac{\omega_0}{2} x^2 \quad \text{--- (4a)}$$

$$V(x) = EI \left(6C_3 + \frac{\omega_0}{EI} x \right)$$

$$\Rightarrow V(x) = \frac{-5}{8} \omega_0 L + \omega_0 x \quad \text{--- (5a)}$$

(b) Pure Shear Beam Theory



Equilibrium:

$$V + w(x)dx = V + dv$$

$$\Rightarrow \frac{dv}{dx} = w(x) \quad \text{--- (a)}$$

$$\frac{dM}{dx} = V \quad \text{--- (a)}$$

Constitutive:

$$V = GA_s \gamma(x) \quad \text{--- (b)}$$

Compatibility:

$$\gamma(x) = \frac{dy}{dx} \quad \text{--- (c)}$$

for constant GA_s ,

$$-GA_s \frac{d^2y}{dx^2} = w(x) \quad \text{--- (d)} \quad \Rightarrow \quad \frac{d^2y}{dx^2} = -\frac{w_0}{GA_s}$$

Integrating eq (d) twice,

$$y(x) = C_4 + C_5 x - \frac{w_0 x^2}{2GA_s} \quad \text{--- (e)}$$

$$\gamma(x) = C_5 - \frac{w_0 x}{GA_s} \quad \text{--- (f)}$$

Boundary Conditions,

$$\text{At } x=0, y(0)=0 \Rightarrow C_4=0 \quad \text{--- (g)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow C_5 L - \frac{\omega_0 L^2}{2GA_s} = 0 \Rightarrow C_5 = +\frac{\omega_0 L}{2GA_s} \quad \text{--- (h)}$$

Thus,

$$y(x) = +\frac{\omega_0 L}{2GA_s} x - \frac{\omega_0 x^2}{2GA_s} \quad \text{--- (1b)}$$

$$\alpha(x)=0 \quad \text{--- (2b)}$$

$$\gamma(x) = -\frac{\omega_0 L}{2GA_s} + \frac{\omega_0 x}{GA_s} \quad \text{--- (3b)}$$

$$M(x) = \int V dx \quad \text{--- (4b)}$$

$$V(x) = -\frac{\omega_0 L}{2} + \omega_0 x \quad \text{--- (5b)}$$

$$\Rightarrow M(x) = -\frac{\omega_0 L}{2} x + \frac{\omega_0 x^2}{2} + C_6$$

$$\text{At } x=L, M(L)=0 \Rightarrow -\frac{\omega_0 L^2}{2} + \frac{\omega_0 L^2}{2} + C_6 = 0 \Rightarrow C_6 = 0$$

$\therefore$ Eqⁿ (4b) becomes,

$$M(x) = -\frac{\omega_0 L}{2} x + \frac{\omega_0 x^2}{2} \quad \text{--- (4b)}$$

(c) Timoshenko Beam Theory

Equilibrium:

$$\frac{dV}{dx} = w(x) \quad \text{--- (1)}$$

$$\frac{dM}{dx} = V(x) \quad \text{--- (2)}$$

Constitutive:

$$K = \frac{1}{\beta} = \frac{M}{EI} \quad \text{--- (3)}$$

$$V(x) = GA_s \gamma(x) \quad \text{--- (4)}$$

Compatibility:

$$K(x) = \frac{d\alpha(x)}{dx} \quad \text{--- (5)}$$

$$\frac{dy(x)}{dx} = \alpha(x) - \gamma(x) \quad \text{--- (6)}$$

$\hookrightarrow$ due to Shear force
 $\rightarrow$ due to bending

$$\Rightarrow \gamma(x) = \alpha(x) - \frac{dy(x)}{dx}$$

from (1), (2), (3) and (5),
for constant EI,

$$EI \frac{d^3 \alpha(x)}{dx^3} = w(x) \quad \text{--- (7)}$$

And, (2), (4), (6) gives,

$$\frac{dy(x)}{dx} = \alpha(x) - \frac{V(x)}{GA_s} = \alpha(x) - \frac{EI}{GA_s} \frac{d^2 \alpha(x)}{dx^2} \quad \text{--- (8)}$$

Integrating eqn (7) three times,

$$\alpha(x) = a_0 + a_1 x + a_2 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (9)}$$

$$\Rightarrow y(x) = a_0 + a_1 x + a_2 x^2 + \frac{w_0}{6EI} x^3 - \frac{EI}{GA_s} \left(2a_2 + \frac{w_0}{EI} x \right)$$

$$\Rightarrow y(x) = \left(a_0 - \frac{2EI a_2}{GA_s} \right) x + \left(a_1 - \frac{w_0}{GA_s} \right) \frac{x^2}{2} + a_2 \frac{x^3}{3} + \frac{w_0}{6EI} \left(\frac{x^4}{4} \right) + a_3 \quad \text{--- (10)}$$

Boundary Conditions;

$$\text{At } x=0, y(0)=0 \Rightarrow a_3=0 \quad \text{--- (11)}$$

$$\text{At } x=0, \alpha(0)=0 \Rightarrow a_0=0 \quad \text{--- (12)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow -\frac{2EI}{GA_s} a_2 L + \left(a_1 - \frac{w_0}{GA_s} \right) \frac{L^2}{2} + a_2 \frac{L^3}{3} + \frac{w_0}{6EI} \frac{L^4}{4} = 0 \quad \text{--- (13)}$$

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$$\text{At } x=L, M(L)=0 \Rightarrow \frac{EI d^2 \alpha(L)}{dx^2} = 0$$

$$\Rightarrow a_1 + 2a_2 L + \frac{\omega_0}{6EI} 3L^2 = 0$$

$$\Rightarrow a_1 = -2a_2 L - \frac{\omega_0 L^2}{2EI} \quad (14)$$

Solving (13) and (14),

$$a_2 = \frac{\frac{\omega_0 L^2}{2GA_s} + \frac{5}{24} \frac{\omega_0 L^4}{EI}}{\frac{2EI L}{GA_s} - \frac{2L^3}{3}} \quad (15)$$

$$a_1 = -2L \left(\frac{\frac{\omega_0 L^2}{2GA_s} + \frac{5}{24} \frac{\omega_0 L^4}{EI}}{\frac{2EI L}{GA_s} - \frac{2L^3}{3}} \right) + \frac{\omega_0 L^2}{2EI} \quad (16)$$

Thus,

$$y(x) = \frac{-2EI}{GA_s} a_2 x + \frac{a_2 x^2}{2} - \frac{\omega_0}{GA_s} \frac{x^2}{2} + \frac{a_2 x^3}{3} + \frac{\omega_0}{6EI} \frac{x^4}{4} \quad (17)$$

$$\alpha(x) = a_1 x + a_2 x^2 + \frac{\omega_0}{6EI} x^3 \quad (18)$$

$$V(x) = \frac{2EI}{GA_s} a_2 + \frac{\omega_0}{GA_s} x \quad (19)$$

$$M(x) = EI a_1 + 2EI a_2 x + \frac{w_0}{2} x^2 \quad \text{--- (4c)}$$

$$V(x) = 2EI a_2 + w_0 x \quad \text{--- (5c)}$$

This derived solution (1a), (2c), (3c), (4c), (5c) yields,

when $GA_s \rightarrow \infty$ i.e. Euler-Bernoulli Beam:

$$a_2 = -\frac{5}{16} \frac{w_0 l}{EI}, \quad a_1 = \frac{1}{8} \frac{w_0 l^2}{EI}$$

$$\therefore y(x) = 0 + \frac{1}{16} \frac{w_0 l^2}{EI} x^2 - 0 - \frac{5}{48} \frac{w_0 l}{EI} x^3 + \frac{w_0}{24EI} x^4$$

$$\alpha(x) = \frac{1}{8} \frac{w_0 l^2}{EI} x - \frac{5}{16} \frac{w_0 l}{EI} x^2 + \frac{w_0}{6EI} x^3$$

$$\gamma(x) = 0$$

$$M(x) = \frac{w_0 l^2}{8} - \frac{5}{8} w_0 l x + \frac{w_0}{2} x^2$$

$$V(x) = -\frac{5}{8} w_0 l + w_0 x$$

which are same as equations (1a), (2a), (3a), (4a) and (5a) respectively.

When $EI \rightarrow \infty$ i.e. pure shear beam,

$$a_2 = 0, a_4 = 0$$

$$2EI a_3 = -\frac{w_0 l}{2}$$

$$EI a_5 = 0$$

$$\therefore y(x) = \frac{w_0 l}{2GA_s} x - \frac{w_0 x^2}{2GA_s}$$

$$\alpha(x) = 0$$

$$\chi(x) = -\frac{w_0 l}{2GA_s} + \frac{w_0 x}{GA_s}$$

$$M(x) = -\frac{w_0 l}{2} x + \frac{w_0 x^2}{2}$$

$$V(x) = -\frac{w_0 l}{2} + w_0 x$$

which are same as equations (1b), (2b), (3b), (4b) and (5b) respectively.

Problem # 3

Consider the following properties for the beam shown in Problem #2:

Steel material: $E = 29000$ ksi, $\nu = 0.3$, $G = \frac{E}{2(1+\nu)}$

Beam section: W18x175, $h = 20.04$ in, $I = 3450$ in⁴, $A_s = 17.8$ in²

Load: $w_0 = 2$ kips/in

Consider three values for length $L = [2, 5, 20]$ ft, plot and compare the vertical deformation $y(x)$, internal shear $V(x)$, and internal moment $M(x)$ for the three cases. Study the differences as a function of h/L and $\phi = \frac{12EI}{GA_s L^2}$ and try to understand the effects of shear stiffness on the beam response. State your observations and conclusions.

Code: https://github.com/nirajkyadav/Homeworks/blob/main/HW1/Q3_updated.py

Observations

1. Vertical Deformation $y(x)$ Comparison

- The Deep Case ($L = 2$ ft, $h/L = 0.835$, $\phi = 10.50$): For a short, deep beam, bending deflection (Euler-Bernoulli) drops to nearly zero, while shear deformation captures the majority of the deflection. The Timoshenko curve sits slightly above the Pure-shear curve, showing that the bending component only adds a minor correction.
- The Intermediate Case ($L = 5$ ft, $h/L = 0.334$, $\phi = 1.68$): For the intermediate beam, both bending and shear contributions are significant. The Timoshenko curve is visibly a combination of both deflections.
- The Slender Case ($L = 20$ ft, $h/L = 0.083$, $\phi = 0.10$): When the beam is long and slender, bending dominates the behavior. The Euler-Bernoulli curve captures the majority of the deflection, while the Pure Shear model significantly underestimates it. The Timoshenko curve sits slightly above the Euler-Bernoulli curve, showing that shear only adds a minor correction.

2. Internal Shear $V(x)$ Comparison

Similar to the deflection, for the short beam internal shear line estimated by Pure Shear Beam Theory is close to that estimated by Timoshenko Beam Theory. The long slender beam internal shear line estimated by Euler-Bernoulli Beam Theory overlaps the Timoshenko Beam Theory line. For the intermediate beam ($L = 5$ ft), the Timoshenko line sits in the middle of the other two.

3. Internal Moment $M(x)$ Comparison

The bending moment at the fixed support estimated by Euler-Bernoulli Beam Theory is significantly higher than that estimated by the Pure Shear and Timoshenko Beam Theory. However, as the beam becomes longer and more slender, the Timoshenko curve shifts upward until it overlaps with the Euler-Bernoulli curve for $L = 20$ ft.

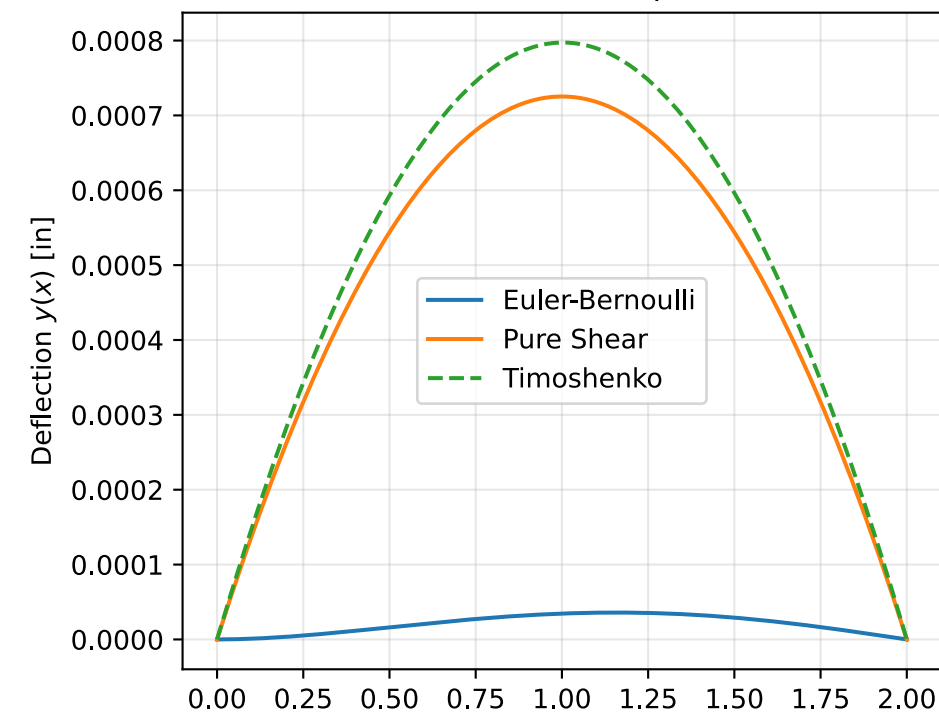
In contrast, Pure Shear Beam Theory predicts 0 moments at the fixed support for all the cases. This is because pure shear theory assumes infinite bending stiffness, and the beam does not experience any bending curvature. Therefore no fixed-end moments are produced.

Conclusions

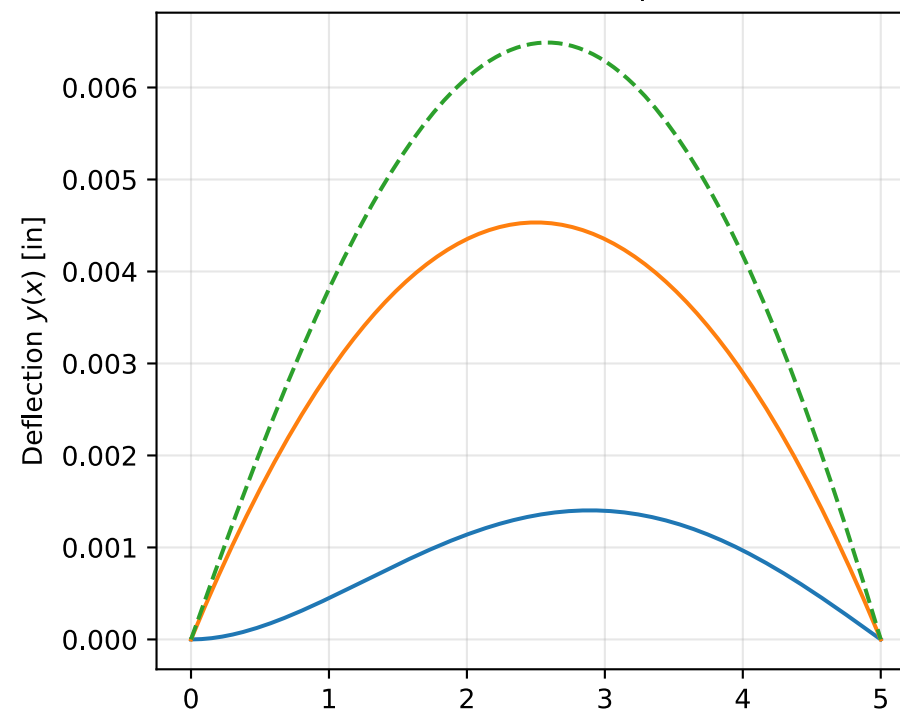
These non-dimensional ratios can predict the behaviour of beams. When ϕ is very small (slender beams), shear forces don't change the shape much, and Euler-Bernoulli theory works estimates the behaviour well. When ϕ is large (short, deep beams), shear deformation can estimate the behaviour well. For intermediate beams both deformations should be considered.

Pure Shear is an unrealistic extreme because it estimates 0 moments at fixed supports for all cases, and Euler-Bernoulli overestimates the moments significantly for deep blocks. Timoshenko beam theory is the general model and can estimate the behaviour in all cases.

L = 2 ft ($h/L=0.835$, $\phi=10.50$)



L = 5 ft ($h/L=0.334$, $\phi=1.68$)



L = 20 ft ($h/L=0.083$, $\phi=0.10$)

